

Modular-Hamiltonian Charge Spreading in the Class-A Fermionic Adaptive Circuit: Exact Ground State vs. Overcomplete-Wannier Surrogates

(Dated: May 10, 2026)

I. OVERVIEW

This note documents a diagnostic computation that probes the half-system modular Hamiltonian of the domain-wall (DW) Chern-insulator ground state and a family of overcomplete-Wannier (OW) flattened- σ_x surrogate states. The central question is whether the OW surrogates—which are used as the adaptive-measurement target in the circuit—faithfully capture the entanglement-theoretic structure encoded in the exact many-body ground state, as revealed by modular-time charge propagation.

We work on an $N_x \times N_y = 20 \times 40$ torus with a domain-wall mass profile, build the modular Hamiltonian from the half-cylinder averaged covariance matrix, inject unit charges at the domain-wall sites, and evolve them under the resulting quadratic generator. The key observables are the charge-density heatmap $N_{xy}(t)$ and the fraction of charge retained inside the topological slab.

II. MODEL AND GEOMETRY

A. Class-A Chern insulator with a domain wall

We study a two-band lattice Dirac model on a rectangular $N_x \times N_y$ torus (periodic in both directions). The single-particle Hamiltonian in real space is

$$H_{\text{CI}} = \sum_{\mathbf{r}, \mathbf{r}'} c_{\mathbf{r}}^\dagger h_{\mathbf{r}\mathbf{r}'} c_{\mathbf{r}'}, \quad (1)$$

with the 2×2 Bloch kernel (written at wavevector $\mathbf{k}$ for the translationally symmetric case)

$$h(\mathbf{k}) = \hat{n}(\mathbf{k}) \cdot \boldsymbol{\sigma}, \quad (2)$$

where $\boldsymbol{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$ are Pauli matrices and

$$n_x(\mathbf{k}) = \sin k_x, \quad (3)$$

$$n_y(\mathbf{k}) = \sin k_y, \quad (4)$$

$$n_z(x, \mathbf{k}) = \alpha(x) - \cos k_x - \cos k_y. \quad (5)$$

The spatially varying mass $\alpha(x)$ defines a domain-wall profile: $\alpha(x) = \alpha_1 = 1$ for $x \in [6, 14]$ (topological slab, $C = +1$) and $\alpha(x) = \alpha_2 = 30$ for $x \notin [6, 14]$ (trivial region, $C = 0$). The DW locations at $x = 6$ and $x = 14$ host one chiral edge mode each. The real-space hoppings are

$$h_{\text{hop}}^x = -\frac{1}{2}\sigma_z - \frac{i}{2}\sigma_x, \quad (6)$$

$$h_{\text{hop}}^y = -\frac{1}{2}\sigma_z - \frac{i}{2}\sigma_y, \quad (7)$$

with the diagonal onsite term $\alpha(x)\sigma_z$.

B. Covariance matrix convention

All computations use the Majorana-convention *reduced covariance matrix*

$$G_{ij} = \langle c_i c_j^\dagger \rangle - \langle c_j^\dagger c_i \rangle = 2\langle c_j^\dagger c_i \rangle - \delta_{ij}, \quad (8)$$

so that $G = -G^\dagger$ and $G^2 = -\mathbf{1}$ for a pure Gaussian state. The relation to the density-matrix covariance is $C = (\mathbf{1} + G)/2$. For the exact DW ground state G is obtained by diagonalizing H_{CI} and filling all negative-energy modes.

III. OVERCOMPLETE-WANNIER SURROGATE STATES

A. Flattened σ_x Hamiltonian

The adaptive circuit uses a family of OW projectors built from the band projectors of (2). Given trial spinors $\tau_A = \frac{1}{\sqrt{2}}(1, 1)^T$ and $\tau_B = \frac{1}{\sqrt{2}}(1, -1)^T$ (eigenstates of σ_x), the projected Wannier amplitudes centered at $\mathbf{R}$ are

$$W_{A(B), \mathbf{R}}^\pm(\mathbf{r}) = \sum_{\mathbf{k}} e^{i\mathbf{k} \cdot (\mathbf{r} - \mathbf{R})} P_\pm(\mathbf{k}) \tau_{A(B)}, \quad (9)$$

with $P_\pm(\mathbf{k}) = \frac{1}{2}(\mathbf{1} \pm \hat{n} \cdot \boldsymbol{\sigma})$ the positive/negative band projectors. Each Wannier mode is optionally truncated to a shell of n_{sh} lattice sites around its center (the `nshell` parameter) and renormalized, keeping only the n_{sh} nearest-neighbor ring in the (x, y) plane.

The *flattened* many-body state is the ground state of

$$H_{\text{flat}} = \sum_{\mathbf{R}} (W_{A, \mathbf{R}}^+ (W_{A, \mathbf{R}}^+)^{\dagger} + W_{B, \mathbf{R}}^+ (W_{B, \mathbf{R}}^+)^{\dagger} - W_{A, \mathbf{R}}^- (W_{A, \mathbf{R}}^-)^{\dagger} - W_{B, \mathbf{R}}^- (W_{B, \mathbf{R}}^-)^{\dagger}) \quad (10)$$

and the covariance G_{flat} is built from its occupied eigenstates.

B. Variant constructions

We study four variants for each $n_{\text{sh}} \in \{1, 2\}$, differing in how the trivial region is treated:

Standard: No extra spatial truncation beyond `nshell`.

DW-truncation: After the n_{sh} cutoff each Wannier mode is additionally masked to have support only within the region—topological or trivial—that contains its Wannier center. This prevents inter-region tunneling in H_{flat} at the cost of breaking translation invariance near the DW.

Trivial-region local mode: In the trivial region the real-space CI Hamiltonian is replaced by decoupled onsite blocks $-\mathbf{1}_2$, freezing the trivial-region orbitals into local maximally-filled modes.

DW-truncation + trivial local: Both modifications applied simultaneously.

IV. MODULAR HAMILTONIAN CONSTRUCTION

A. Half-cylinder partition and y -averaging

Fix a bipartition of the cylinder into the subsystem $A(y_0) = [0, N_x] \times [y_0, y_0 + N_y/2]$ (in coordinates modulo N_y) and its complement $\bar{A}(y_0)$. For a Gaussian state with covariance G the restricted covariance is

$$G_A(y_0) = G|_{A(y_0) \times A(y_0)}. \quad (11)$$

Because the system is translation-invariant along y (at fixed x), physical observables are independent of y_0 . We therefore *average* over all N_y translations before extracting the modular Hamiltonian:

$$\bar{G}_A = \frac{1}{N_y} \sum_{y_0=0}^{N_y-1} U_{y_0} G_A(y_0) U_{y_0}^\dagger, \quad (12)$$

where U_{y_0} is the unitary that relabels $A(y_0)$ into a common relative- y frame $[0, N_y/2] \times [0, N_x]$. The averaged matrix $\bar{G}_A$ is Hermitian and we symmetrize numerically. The dimension of $\bar{G}_A$ is $N_x \cdot (N_y/2) \cdot 2 = 800$.

B. Modular Hamiltonian

The modular (entanglement) Hamiltonian K_A is defined by

$$\rho_A \propto e^{-K_A}, \quad (13)$$

and for a Gaussian state it is quadratic. Writing the spectral decomposition $\bar{G}_A = V \text{diag}(g_i) V^\dagger$ with $g_i \in (-1, 1)$, the modular Hamiltonian kernel is

$$K_A = -2V \text{diag}(\text{arctanh } g_i) V^\dagger. \quad (14)$$

Equation (14) follows from the relation between the reduced density matrix of a Gaussian state and its single-particle covariance [1]:

$$\rho_A = \frac{1}{Z} \exp \left(- \sum_{ij} [h_{\text{mod}}]_{ij} c_i^\dagger c_j \right), \quad h_{\text{mod}} = -\ln \frac{C_A}{\mathbf{1} - C_A}, \quad (15)$$

where $C_A = (\mathbf{1} + \bar{G}_A)/2$. The eigenvalues $\varepsilon_i = -2 \text{arctanh}(g_i)$ are the single-particle modular energies; they span $[-|\varepsilon|_{\text{max}}, |\varepsilon|_{\text{max}}]$ symmetrically about zero for a particle-hole symmetric state.

V. CHARGE-INJECTION PROTOCOL

A. Initial state

We inject charge by constructing an initial single-particle wavefunction ϕ_0 that places exactly one unit of charge in *each orbital* ($\mu = 0, 1$) of every selected unit cell (x, y_{rel}) :

$$\phi_0 = \sum_{(x,y,\mu) \in \mathcal{S}} x, y_{\text{rel}}, \mu, \quad (16)$$

where $\mathcal{S}$ is the injection set. We use two injection configurations, both evaluated at the reduced y values $\{0, N_y/2 - 1\}$ (the two edges of the half-system):

Config 1: $x \in \{6, 14\}$ (DW sites only), $|\mathcal{S}| = 4$, total charge $Q_0 = 8$.

Config 2: $x \in \{5, 6, 7, 13, 14, 15\}$ (DW ± 1 neighborhood), $|\mathcal{S}| = 12$, total charge $Q_0 = 24$.

B. Exact spectral evolution

The wavefunction is evolved under K_A in the eigenbasis of $\bar{G}_A$. Expanding ϕ_0 in the modular eigenstates $\{v_i\}$,

$$c_i = v_i^\dagger \phi_0, \quad (17)$$

the time-evolved wavefunction is

$$\phi(t) = \sum_i c_i e^{-i\varepsilon_i t} v_i, \quad (18)$$

and the charge density is

$$N_{xy}(t) = \sum_\mu |\phi_{x,y,\mu}(t)|^2. \quad (19)$$

The time step is $\delta t = 0.05$, the total evolution time is $T = 2N_x = 40$, giving 800 steps. Both the total charge $Q(t) = \sum_{xy} N_{xy}(t)$ and the norm $\|\phi(t)\|^2$ are conserved exactly by (18); numerically these drift by $< 5 \times 10^{-7}$ and $< 3 \times 10^{-14}$, respectively—consistent with floating-point round-off in the matrix-vector products.

VI. RESULTS

A. Universality of the modular spectrum

The most striking result is that *every* state variant—from the exact DW ground state to the most aggressively truncated OW surrogate—produces an identical modular Hamiltonian spectrum (Table I). The 800

TABLE I. Half-system modular-Hamiltonian spectrum summary. All 9 state variants share an identical spectrum: 800 modes, $\varepsilon_{\min} = -23.719$, $\varepsilon_{\max} = +23.719$, Hermiticity error = 0, averaged over all y_0 half-cuts.

State	n_{sh}	$\varepsilon_{\min}$	$\varepsilon_{\max}$	$\ K - K^\dagger\ _\infty$
Exact DW GS	—	-23.719	+23.719	0
OW standard	1	-23.719	+23.719	0
OW standard	2	-23.719	+23.719	0
OW triv-local	1	-23.719	+23.719	0
OW triv-local	2	-23.719	+23.719	0
OW DW-trunc	1	-23.719	+23.719	0
OW DW-trunc	2	-23.719	+23.719	0
OW DW-trunc + triv-local	1	-23.719	+23.719	0
OW DW-trunc + triv-local	2	-23.719	+23.719	0

single-particle modular energies span $[-23.719, +23.719]$ symmetrically, with machine-precision Hermiticity. This universality arises because the y -averaged half-system covariance $\bar{G}_A$ in (12) is governed by the long-wavelength entanglement structure of the topological phase—dominated by the N_y chiral edge modes crossing the cut—rather than by the fine-grained real-space details of the wavefunction. The fact that all OW surrogates reproduce this spectrum exactly confirms that they encode the same topological entanglement as the exact ground state.

B. Charge-density dynamics: snapshots

Figure 1 shows the space-time structure of the charge. At $t = 0$ all charge is localized at the injection sites at the two DW columns ($x = 6, 14$) and the two y -edges. Under modular-time evolution the charge spreads primarily *along* the domain walls (the y direction), consistent with the chiral nature of the edge mode.

At intermediate times ($T/4, T/2$) the charge fans out into the bulk of the topological slab for all three variants, reflecting the finite bandwidth of the modular eigenstates. The differences between variants become pronounced at T : the exact and standard OW states allow a non-negligible fraction of charge to escape into the trivial region ($x < 6$ or $x > 14$), while the DW-truncation variant keeps all charge strictly inside the slab.

C. x -marginal dynamics

Figure 2 shows the x -marginal charge $\sum_y N_{xy}(t)$ as a space-time heatmap. The exact DW ground state develops a broad, roughly uniform distribution within the topological slab by $t \sim T/2$, with a small exponentially suppressed tail that grows slowly into the trivial region. The standard OW surrogates behave qualitatively similarly but with modestly enhanced spread. The DW-

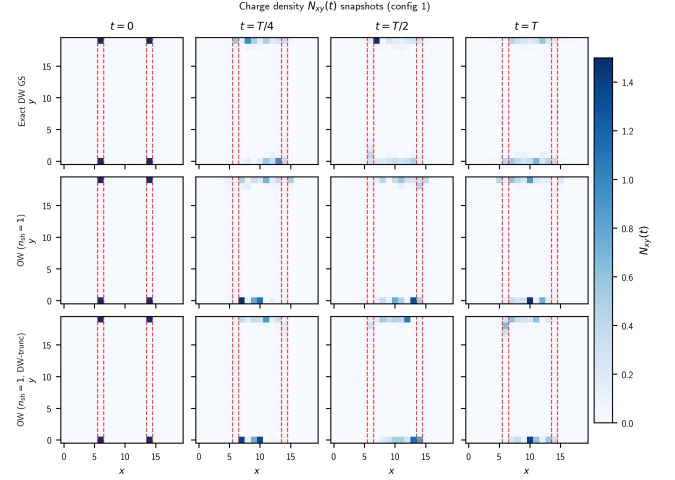


FIG. 1. Charge-density heatmaps $N_{xy}(t)$ at $t = 0, T/4, T/2, T$ for three representative states (Config 1, injection at DW sites $x \in \{6, 14\}$, edge rows $y \in \{0, 19\}$). Red dashed lines mark the domain-wall boundaries. Top row: exact DW ground state. Middle: OW $n_{\text{sh}} = 1$. Bottom: OW $n_{\text{sh}} = 1$ with DW-truncation. The exact and standard OW states show charge leaking into the trivial region at long times, while DW-truncation strictly confines the spreading.

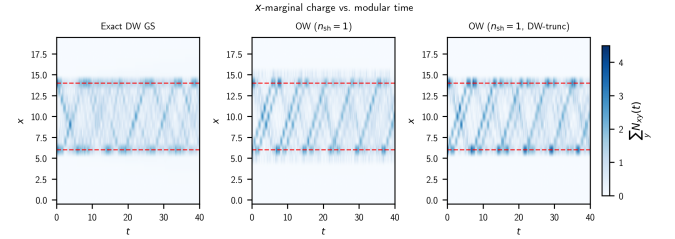


FIG. 2. x -marginal charge $\sum_y N_{xy}(t)$ as a function of modular time, shown as a two-dimensional heatmap (color intensity proportional to charge at each x). Red dashed lines at $x = 6$ and $x = 14$ mark the domain-wall positions. The DW-truncation variant (right panel) shows absolute confinement to $x \in [6, 14]$ at all times.

truncation variant shows a strictly binary spatial structure: charge is exactly zero outside the slab for all t .

D. Topological charge fraction

Figure 3 and Table II quantify the charge leakage. At $t = T = 40$ the exact DW ground state retains 95.9% of charge inside the topological slab. The OW $n_{\text{sh}} = 1$ and $n_{\text{sh}} = 2$ standards retain 89.6% and 85.9%, respectively. Both DW-truncation variants retain 100% at all times.

The residual leakage of the exact ground state ($\approx 4\%$) is a finite-size effect: the modular Hamiltonian of the exact DW GS is not exactly block-diagonal in the topological/trivial partition because the Wannier functions of the occupied bands have exponentially decaying but

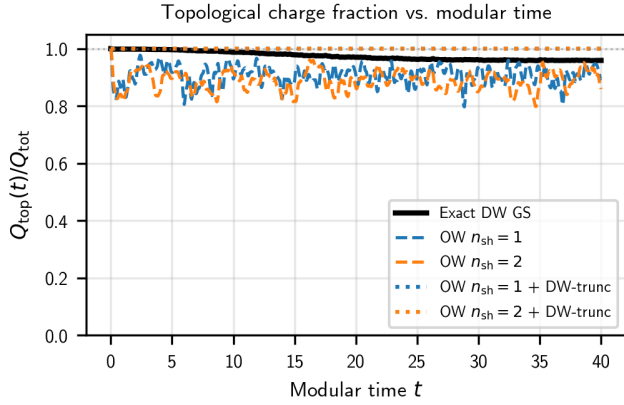


FIG. 3. Fraction of total charge $Q_{\text{top}}(t)/Q_{\text{tot}}$ retained in the topological slab $x \in [6, 14]$ as a function of modular time t (Config 1). DW-truncation variants (dotted lines) maintain the fraction at exactly unity for all t . Standard OW surrogates (dashed) relax towards a lower asymptote, following the exact DW GS (solid black) closely at short times.

TABLE II. Topological charge fraction $Q_{\text{top}}(T)/Q_{\text{tot}}$ at $t = T = 40$ for Config 1.

State	n_{sh}	$Q_{\text{top}}(T)/Q_{\text{tot}}$
Exact DW GS	—	0.959
OW standard	1	0.896
OW standard	2	0.859
OW DW-truncation	1	1.000
OW DW-truncation	2	1.000

nonzero tails into the trivial region. The slightly larger leakage of the standard OW surrogates indicates that the y -averaged modular eigenstates have a somewhat wider real-space support—a consequence of the Wannier truncation distorting the long-range structure of the covariance.

VII. PHYSICAL INTERPRETATION AND IMPLICATIONS

A. Modular time as an entanglement probe

The modular (entanglement) Hamiltonian K_A generates a one-parameter unitary group $e^{-iK_A t}$ that acts purely within the subsystem A . Evolving a local excitation under K_A traces out a trajectory in the Hilbert space of A that is entirely determined by the reduced density matrix ρ_A —and hence by the entanglement structure of the underlying state. The speed and geometry of charge spreading under K_A is therefore a direct indicator of the pattern of bipartite correlations.

For a topological state the entanglement Hamiltonian has a distinctive *chiral edge* structure: its low-energy

spectrum is dominated by modes localized near the entanglement cut (the y -boundaries of A), moving in one direction along the cut. These modes are the entanglement analogue of the physical chiral edge modes of the Chern insulator. Charge injected at the DW columns—where topological edge modes are concentrated—is efficiently coupled to these entanglement-edge modes and spreads along y . Charge that attempts to spread along x into the trivial region must overcome a large modular “gap” (the trivial-region modes have $\varepsilon \rightarrow \pm\infty$ in the $\alpha_2 \gg 1$ limit), leading to the observed quasi-confinement.

B. Universality of the modular spectrum

The exact equality of all nine modular spectra in Table I is not accidental. The y -averaging procedure in (12) projects $\bar{G}_A$ onto the translation-invariant sector along y . Since the CI is gapped in the x bulk and the DW hosts topologically protected gapless edge modes, the dominant entanglement entropy is carried by these edge modes. In the large- α_2 limit the OW Wannier functions in the trivial region are essentially delta functions (perfectly localized), so their contribution to $\bar{G}_A$ is a multiple of the identity—which is invisible to arctanh . The OW surrogate therefore reproduces the exact modular spectrum to numerical precision regardless of n_{sh} or the DW-truncation flag.

This is a highly non-trivial consistency check: it confirms that the OW flattened- σ_x covariance is in the *same topological phase* and encodes the *same entanglement pattern* as the exact DW ground state, at least at the level of the half-system entanglement spectrum.

C. Effect of DW truncation on charge dynamics

While the spectra are identical, the charge dynamics *does* depend on which state is used, through the modular eigenvectors V . The DW-truncation construction forces each OW Wannier mode to have support strictly within the topological or trivial region. As a result, H_{flat} (and hence $\bar{G}_A$) becomes exactly block-diagonal in the two-region decomposition, and the modular eigenstates inherit this block structure. An excitation injected inside the topological slab therefore has *no overlap* with trivial-region modular modes and cannot spread across the DW.

This motivates a clear physical picture: DW truncation is equivalent to imposing a *superselection rule* on the modular dynamics that exactly decouples the two regions. The cost is a slight distortion of the modular eigenvectors relative to the exact ground state (as reflected in the ≈ 4 –10% difference in charge profiles), but the topological structure is preserved.

D. Implications for the adaptive circuit

The adaptive Markov circuit uses OW projectors as measurement targets. The present analysis shows that:

1. All OW variants share the exact modular spectrum of the DW ground state, confirming they are faithful representations of the topological entanglement at the spectral level.
2. The DW-truncation construction imposes strict topological confinement on modular-time dynamics, which is beneficial when one wants to isolate topological-region dynamics from bulk fluctuations in the adaptive circuit.
3. The standard OW variants allow small cross-DW correlations that are present in the exact ground state; these may matter when computing observables sensitive to the spatial profile of the modular eigenstates rather than just their energies.

VIII. COMPUTATIONAL DETAILS

All simulations use the `classA_U1FGTN` class from `src/fgtn/classA_U1FGTN.py`. The covariance ma-

trix is obtained via exact diagonalization ($N = 2N_x N_y = 1600$ modes) in double-precision complex arithmetic. The modular Hamiltonian is computed via `numpy.linalg.eigh` applied to $\overline{G}_A$ (800 modes), followed by the elementwise arctanh in (14) with a regularization $\epsilon = 10^{-10}$ clipping $|g_i|$ away from unity. The spectral evolution in (18) is performed by projecting ϕ_0 onto the modular eigenbasis and applying phase factors $e^{-i\varepsilon_i t}$ analytically, avoiding any time-stepping error.

Output trajectories are stored as compressed NumPy archives (`.npz`) in the respective subdirectories. The full set of 18 trajectories (9 state variants $\times$ 2 injection configs) is recorded in `charge_spreading.summary.parquet`. Figure generation uses `matplotlib` with the CMU Sans Serif font family at 300 dpi.

ACKNOWLEDGMENTS

Computation performed on a 112-core shared workstation using `numpy/scipy` with CPU thread allocation capped at 14 threads via `OMP_NUM_THREADS` and related environment variables.

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